Process Technology:
TSMC CL013G

High-Speed Dual-Port Synchronous SRAM

ram_dp_128x12
128X12, Mux 4, Drive 6

Features

- Precise Optimization for TSMC's Eight-Layer Metal 0.13µm CL013G CMOS Process
- High Density (area is 0.037mm^2)
- Fast Access Time (0.98ns at typical process, 1.20V, 25˚C)
- Fast Cycle Time (0.90ns at typical process, 1.20V, 25˚C)
- Two Read/Write Ports
- Completely Static Operation
- Near-Zero Hold Time (Data, Address, and Control Inputs)

Memory Description

The 128X12 SRAM is a high-performance, synchronous dual-port, 128-word by 12-bit memory designed to take full advantage of TSMC's eight-layer metal, 0.13µm CL013G CMOS process.

The SRAM's storage array is composed of eight-transistor cells with fully static memory circuitry. The SRAM operates at a voltage of 1.2V ± 10% and a junction temperature range of -40˚C to +125˚C.

Port Description

Port	Description
A	Read and Write Port
B	Read and Write Port

Pin Description

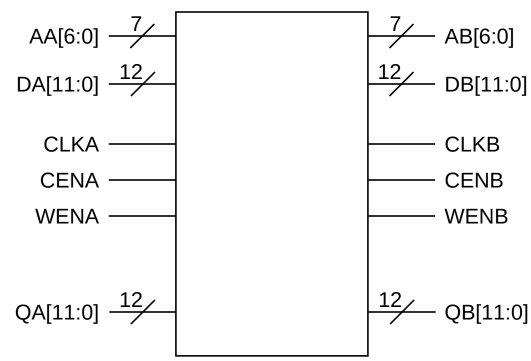
Pin	Description
AA[6:0]	Addresses (AA[0] = LSB)
AB[6:0]	Addresses (AB[0] = LSB)
DA[11:0]	Data Inputs (DA[0] = LSB)
DB[11:0]	Data Inputs (DB[0] = LSB)
CLKA, CLKB	Clock Inputs
CENA, CENB	Chip Enables
WENA, WENB	Write Enables
QA[11:0]	Data Outputs (QA[0] = LSB)
QB[11:0]	Data Outputs (QB[0] = LSB)

Area

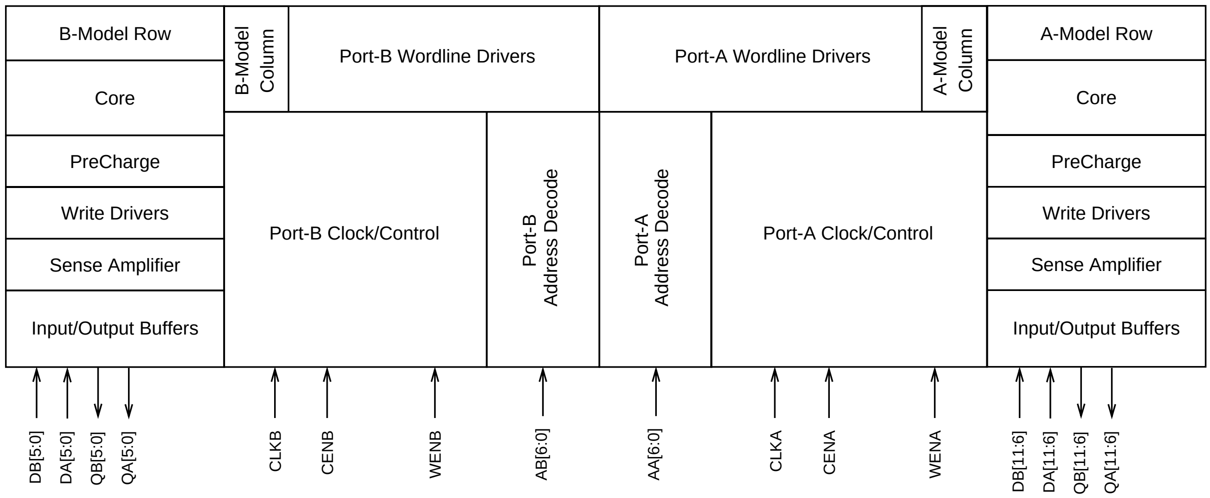
Area Type	Width (mm)	Height (mm)	Area (mm^2)
Core	0.236	0.158	0.037
Footprint	0.246	0.168	0.041

The footprint area includes the core area and user-defined power ring and pin spacing areas.

Symbol

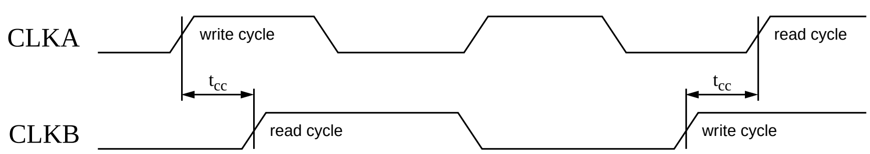


SRAM Block Diagram



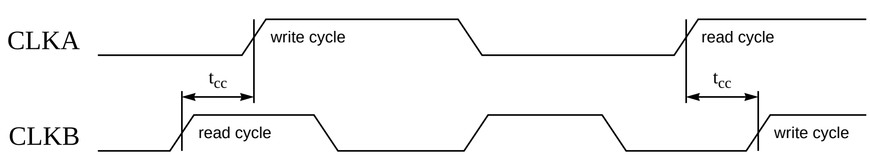
Clock Timing

Figure 1. Synchronous Dual-Port SRAM Write-Read Clock Timing (Accessing Same Address)



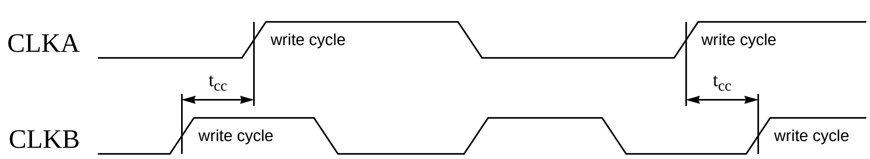
Rising delays are measured at 50% of VDD and falling delays are measured at 50% of VDD.
Rising and falling slews are measured from 10% VDD to 90% VDD.

Figure 2. Synchronous Dual-Port SRAM Read-Write Clock Timing (Accessing Same Address)



Rising delays are measured at 50% of VDD and falling delays are measured at 50% of VDD.
Rising and falling slews are measured from 10% VDD to 90% VDD.

Figure 3. Synchronous Dual-Port SRAM Write-Write Clock Timing (Accessing Same Address)



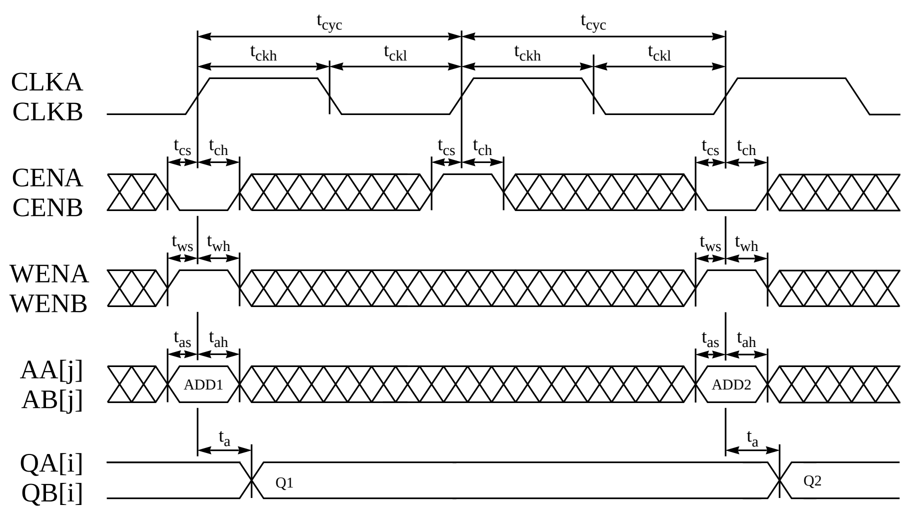
Rising delays are measured at 50% of VDD and falling delays are measured at 50% of VDD.
Rising and falling slews are measured from 10% VDD to 90% VDD.

Read and Write Behavior When Accessing Same Address

Action	Condition	Behavior
write from one port then read from the other port	t_{cc} is satisfied (see Figure 1)	write OK D-to-Q write through OK read (new data) OK
	t_{cc} is not satisfied (see Figure 1)	write OK D-to-Q write through OK read fails
read from one port then write from the other port	t_{cc} is satisfied (see Figure 2)	write OK D-to-Q write through OK read (old data) OK
	t_{cc} is not satisfied (see Figure 2)	write OK D-to-Q write through OK read fails
write from one port then write from the other port	t_{cc} is satisfied (see Figure 3)	both writes OK (second write overwrites first write) D-to-Q write through OK
	t_{cc} is not satisfied (see Figure 3)	both writes fail D-to-Q write through OK
read from one port then read from the other port	no restriction	both reads OK

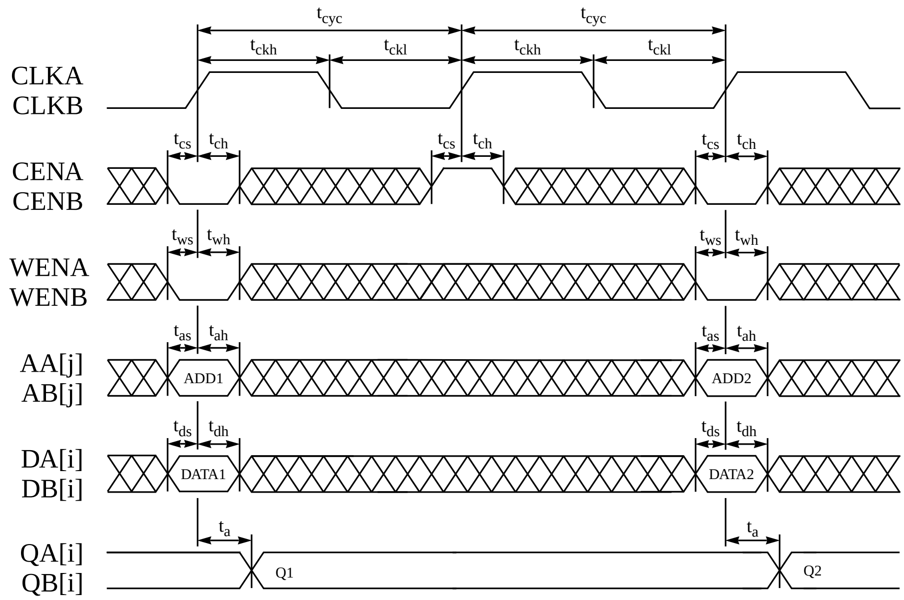
Mission Mode

Figure 4. Synchronous Dual-Port SRAM Read-Cycle Timing



Rising delays are measured at 50% of VDD and falling delays are measured at 50% of VDD.
Rising and falling slews are measured from 10% VDD to 90% VDD.

Figure 5. Synchronous Dual-Port SRAM Write-Cycle Timing



Rising delays are measured at 50% of VDD and falling delays are measured at 50% of VDD.
Rising and falling slews are measured from 10% VDD to 90% VDD.

SRAM Logic Table

CENA	CENB	WENA	WENB	Data Out	Mode	Function
H	X	X	X	Last Data	Port-A Standby	Port-A address inputs are disabled; data stored in the memory is retained, but port A cannot be accessed for new reads or writes. Port-A data outputs remain stable.
X	H	X	X	Last Data	Port-B Standby	Port-B address inputs are disabled; data stored in the memory is retained, but port B cannot be accessed for new reads or writes. Port-B data outputs remain stable.
L	X	L	X	Data In	Port-A Write	Data on the port-A data input bus DA[n-1:0] is written to the memory location specified on the port-A address bus AA[m-1:0], and driven through to the port-A data output bus QA[n-1:0].
X	L	X	L	Data In	Port-B Write	Data on the port-B data input bus DB[n-1:0] is written to the memory location specified on the port-B address bus AB[m-1:0], and driven through to the port-B data output bus QB[n-1:0].
L	X	H	X	SRAM Data	Port-A Read	Data on the port-A data output bus QA[n-1:0] is read from the memory location specified on the port-A address bus AA[m-1:0].
X	L	X	H	SRAM Data	Port-B Read	Data on the port-B data output bus QB[n-1:0] is read from the memory location specified on the port-B address bus AB[m-1:0].

SRAM Timing: Mission Mode

Parameter	Symbol	Fast@-40C Process 1.32V, -40˚C		Typical Process 1.20V, 25˚C		Slow Process 1.08V, 125˚C		Fast@0C Process 1.32V, 0˚C	
		Min (ns)	Max (ns)	Min (ns)	Max (ns)	Min (ns)	Max (ns)	Min (ns)	Max (ns)
Cycle time	t_{cyc}	0.59		0.90		1.52		0.63	
Access time[1,2,3]	t_a	0.60			0.98		1.59	0.63	
Address setup	t_{as}	0.17		0.29		0.54		0.19	
Address hold	t_{ah}	0.00		0.00		0.00		0.00	
Chip enable setup	t_{cs}	0.26		0.39		0.62		0.28	
Chip enable hold	t_{ch}	0.00		0.00		0.00		0.00	
Write enable setup	t_{ws}	0.31		0.45		0.67		0.32	
Write enable hold	t_{wh}	0.00		0.00		0.00		0.00	
Data setup	t_{ds}	0.16		0.26		0.41		0.16	
Data hold	t_{dh}	0.00		0.00		0.00		0.00	
Clock high	t_{ckh}	0.04		0.06		0.10		0.04	
Clock low	t_{ckl}	0.11		0.17		0.28		0.11	
Clock rise slew	t_{ckr}		4.00		4.00		4.00		4.00
Clock collision	t_{cc}	0.36		0.55		0.91		0.38	
Output load factor (ns/pF)	K_{load}		0.55		0.76		1.13		0.56

[1] Parameters have a load dependence (K_{load}), which is used to calculate: $TotalDelay = FixedDelay + (Kload \times Cload)$.

[2] Access time is defined as the slowest possible output transition for the typical and slow corners, and the fastest possible output transition for the fast corner.

Pin Capacitance

Pin	Fast@-40C Process 1.32V, -40˚C	Typical Process 1.20V, 25˚C	Slow Process 1.08V, 125˚C	Fast@0C Process 1.32V, 0˚C
	Value (pF)	Value (pF)	Value (pF)	Value (pF)
AA[j:0], AB[j:0]	0.019	0.018	0.017	0.019
DA[i:0], DB[i:0]	0.002	0.002	0.001	0.002
CLKA, CLKB	0.180	0.171	0.164	0.182
CENA, CENB	0.012	0.011	0.011	0.012
WENA, WENB	0.011	0.011	0.011	0.011

Power

1.00MHz Operation

Condition	Fast@-40C Process 1.32V, -40˚C	Typical Process 1.20V, 25˚C	Slow Process 1.08V, 125˚C	Fast@0C Process 1.32V, 0˚C
	Value (mA/port)	Value (mA/port)	Value (mA/port)	Value (mA/port)
AC Current[1]	0.025	0.029	0.067	0.042
Read AC Current	0.022	0.026	0.064	0.038
Write AC Current	0.029	0.032	0.071	0.046
Peak Current	113.342	63.876	36.862	107.625
Deselected Current[2]	0.008	0.014	0.054	0.024
Standby Current[3]	0.003	0.009	0.050	0.019

[1] Value assumes 50% read and write operations.

[2] Value assumes SRAM is deselected, all addresses switch, and 50% of input pins switch. The logic-switching component of deselected power becomes negligibly small if the input pins are held stable by externally controlling these signals with chip select.

[3] Value is independent of frequency and assumes all inputs and outputs are stable.

Clock Noise Limit

Signal	Fast@-40C Process 1.32V, -40˚C		Typical Process 1.20V, 25˚C		Slow Process 1.08V, 125˚C		Fast@0C Process 1.32V, 0˚C	
	Pulse Width (ns)	Voltage (V)	Pulse Width (ns)	Voltage (V)	Pulse Width (ns)	Voltage (V)	Pulse Width (ns)	Voltage (V)
CLKA/B	10.00	0.61	10.00	0.59	10.00	0.55	10.00	0.60

The clock noise limit is the maximum CLK voltage allowable for the indicated pulse width without causing a spurious memory cycle or other memory failure.

Power and Ground Noise Limit

Signal	Fast@-40C Process 1.32V, -40˚C	Typical Process 1.20V, 25˚C	Slow Process 1.08V, 125˚C	Fast@0C Process 1.32V, 0˚C
	Voltage (V)	Voltage (V)	Voltage (V)	Voltage (V)
Power	0.13	0.12	0.11	0.13
Ground	0.13	0.12	0.11	0.13

The power/ground noise limit is the maximum supply voltage transition allowable without causing a memory failure.